Metal Stamping Simulator Development Journal

Advanced Programming AI Project
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Today, I extended my dashboard with simulated production. The updated code produces one item every 2 seconds while running and gives each item a 10% chance of being defective. My teammate's graphics are still a future integration task.

Previous Session

I already had a 1200 by 700 raylib window, a simulator placeholder, production counters and Start, Pause and Reset buttons. Mouse click handling, RUNNING and STOPPED labels, and the success rate calculation were already in the code. The counters used sample values of 100, 90 and 10.

Work Completed

I changed the starting counters to zero and added a production timer. While the simulation is running, the timer accumulates elapsed frame time. Each completed 2-second cycle increases the total and either the good or defective counter. The existing success rate calculation now uses these changing values.

I connected Pause and Reset to the timer. Pause keeps the current cycle progress so Start can resume it. Reset stops production and clears all counters and the timer. I also added cycle progress text to the dashboard.

What I Learned

I learned that elapsed time is more useful than counting frames for a production timer. I subtract 2 seconds after each product so leftover time is kept. A while loop handles more than one completed cycle after a slow frame. I also avoid counting stopped time on the Start or resume frame.

I use a random integer from 1 to 100. Values from 1 to 10 mean defective, and values from 11 to 100 mean good. A 10% chance does not guarantee exactly 10 defects in every 100 products. The total should always equal good products plus defective products.

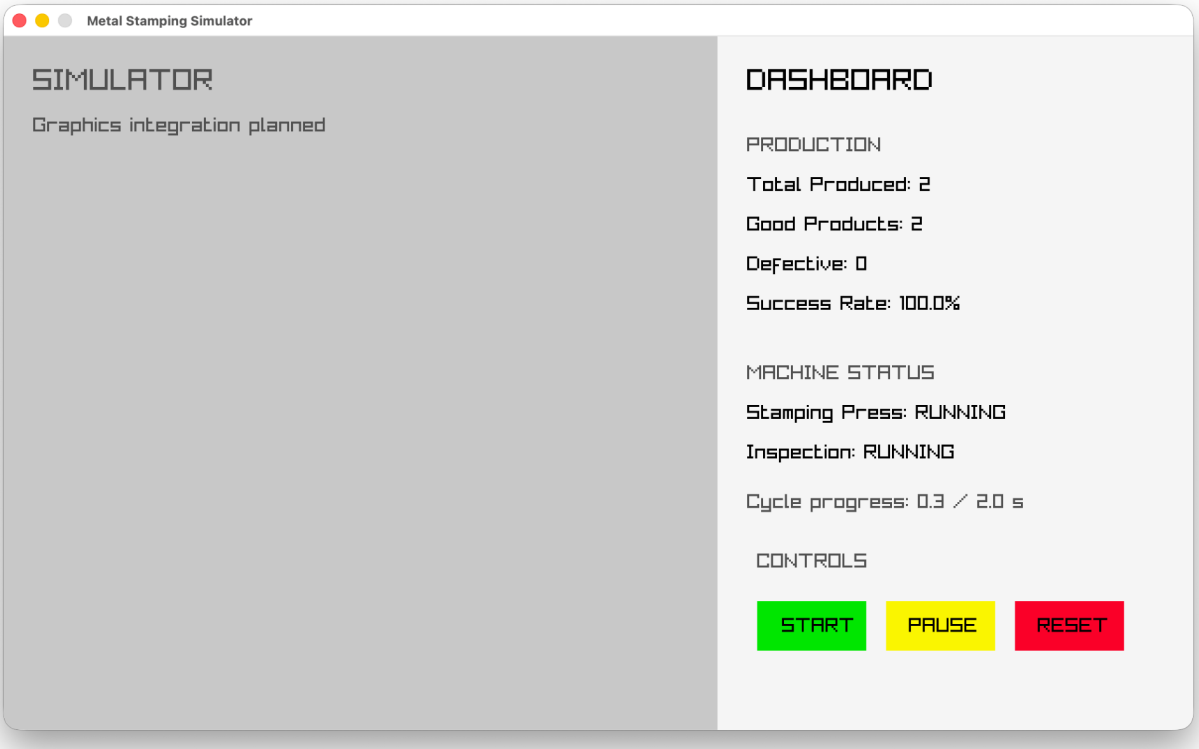
Problems and Checks

The updated code compiled successfully. Automated logic checks passed for timing, pause and resume, reset, defect boundaries, counter totals and success rate. The live window has not yet been checked visually. The left side is still a placeholder, and the random defects are a temporary model rather than a real inspection process.

Next Steps

I will run the program and check the buttons, cycle progress and counters on screen. Then I will confirm my teammate's graphics format and connect the dashboard to the simulator's production and inspection events.

Current Prototype Screenshot


Metal Stamping Simulator
SIMULATOR
Graphics integration planned
DASHBOARD
PRODUCTION
Total Produced: 2
Good Products: 2
Defective: 0
Success Rate: 100.0%
MACHINE STATUS
Stamping Press: RUNNING
Inspection: RUNNING
Cycle progress: 0.3 / 2.0 s
CONTROLS
START
PAUSE
RESET